

2.3.1. Student centric methods, such as experiential learning, participative learning and problem solving methodologies are used for enhancing learning experiences and teachers use ICT- enabled tools including online resources for effective teaching and learning process

The entire teaching-learning process at CSIT is student Centric. Keeping this in view the Institute has incorporated vivid academic aspects in its teaching system. The academic departments look after the academic teaching-learning & co-curricular activities under the leadership of the Director. Apart from the regular classes correlating to the prescribed syllabus, the Departments organize- Guest Lectures, Add-on Courses & Short-term Courses to enable students to become academically strong. We use Practical learning, participative learning, & problem-solving approaches to impart deeper understanding of the subjects.

Experiential Learning Practices

The CSIT has incorporated several experiential learning practices like:

1. Project Works: where the students work in groups on specific areas under the guidance of a faculty member.
2. Learning Through Peers is facilitated through Guest lectures, Seminars & Workshops of the professionals. The students get exposure to the practicability of their acquired knowledge.
3. Activities of Clubs, Cells & Student Associations are organized to enhance managerial skills & imbibe professionalism.
4. Industrial Visits/Tours are coordinated to give exposure to the industrial atmosphere, latest technology & applications of the learnt concepts.
5. Vocational Trainings are done by the students in their respective fields for a period of three to four weeks.
6. Internships: Internships are facilitated to help the students get upper edge thereby exposing them to authentic industrial/workplace demands.

Problem-Solving Based Learning Methods

Problem-Solving based Learning helps the students to learn & get their knowledge enhanced by solving issues, problems & cases. The CSIT exerts following measures in this context:

1. Assignments: Assignments based on real-world challenges are given to the students so as to apply their acquired knowledge.



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2. Presentations: Students' presentations on various topics are organized to improve their communication skills & logical thinking.
3. Tutorials are scheduled where the students solve problems & the teachers guide them to solve the challenging questions they identify.

Participative Learning

CSIT encourages students to participate in all the curricular, extracurricular & academic activities. The Techfest, 'LaQshya' is totally managed by the students.

1. Group learning: Students work in groups on project works assigned to them. The students at CSIT work in groups at hackathon.
2. Team Presentation: After completing project/ Vocational Training & Industrial Visits the students present their views / learnings in team before a designated group of teachers.
3. Participation in Competitions: Students are encouraged & facilitated to actively participate in intercollege / university/zonal events & competitions.

Self Learning

The Institute also facilitates its students to learn by themselves too, realizing the importance of individuality in learning & vividness in learning attitudes. Following are the measures of the CSIT that facilitates for self learning:

1. Central Library: The Central Library is a huge structure that hosts enough titles & a good number of books, magazines & journals. It has sufficient space & comfortable environment for the students to sit & study.
2. e-library: The institutes e-library has internet connection & subscription to e-journals & databases like infolibnet, DELNET etc.



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2.3.1 Student centric methods, such as experiential learning, participative learning and problem solving methodologies are used for enhancing learning experiences

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Teachers use ICT enabled tools for effective teaching- learning Process		
1	Chalk and Board	LINK
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5	Course Hand Out	LINK



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CSIT DURG VARIOUS DEPARTMENT LABS



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Long 81.260034°
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GPS Map Camera



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CSIT DURG VARIOUS DEPARTMENT LABS



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S.No.	Programme name	Program Code	List of students undertaking internship
31	UG-Computer Science and Engineering	BJ7196	AMRIT PATHAK
32	UG-Computer Science and Engineering	BJ7197	RAJENDRA KUMAR
33	UG-Computer Science and Engineering	BJ7198	PRIYANSHU SAHU
34	UG-Computer Science and Engineering	BJ7199	KUMUDINI SAHU
35	UG-Computer Science and Engineering	BJ7200	PRACHI DESHMUKH
36	UG-Computer Science and Engineering	BJ7201	KOMALKANT DESHMUKH
37	UG-Computer Science and Engineering	BJ7219	THANENDRA DESHMUKH
38	UG-Computer Science and Engineering	BJ7220	YAMAN CHANDRA
39	UG-Computer Science and Engineering	BJ7221	JAY KUMAR GUPTA
40	UG-Computer Science and Engineering	BJ7222	BHEDPRAKASH
41	UG-Computer Science and Engineering	BJ7223	YASHVARDHAN PATEL
42	UG-Computer Science and Engineering	BJ7224	LOKESH KUMAR
43	UG-Computer Science and Engineering	BJ7225	SANNIDHYA SAHU
44	UG-Computer Science and Engineering	BJ7226	SHRIKANT TAMRAKAR
45	UG-Computer Science and Engineering	BJ7228	YOGESH KUMAR PATEL
46	UG-Computer Science and Engineering	BJ7230	TANMAY UMAKANT YADAV
47	UG-Computer Science and Engineering	BJ7232	HIMANSHU SONKAR
48	UG-Computer Science and Engineering	BJ7233	SURAJ
49	UG-Computer Science and Engineering	BJ7234	REENA SINHA
50	UG-Computer Science and Engineering	BJ7235	TUSHAR SAHU
51	UG-Computer Science and Engineering	BJ7236	KAJAL VERMA
52	UG-Computer Science and Engineering	BJ7238	NIKITA SONI
53	UG-Computer Science and Engineering	BJ7240	VIKASH RAI
54	UG-Computer Science and Engineering	BJ7241	DEEPAK KUMAR
55	UG-Computer Science and Engineering	BJ7245	R VISHAL REDDY
56	UG-Computer Science and Engineering	BJ7248	DUMESHWARI
57	UG-Computer Science and Engineering	BJ7252	SURAJ PRAKASH PATEL
58	UG-Computer Science and Engineering	BJ7253	MOHAMMAD DANISH
59	UG-Computer Science and Engineering	BJ7254	ARYAN KURIL
60	UG-Computer Science and Engineering	BJ7255	PRASHANT YADAV
61	UG-Computer Science and Engineering	BJ7256	KULDEEP
62	UG-Computer Science and Engineering	BJ7258	ANMOL SINGH
63	UG-Computer Science and Engineering	BJ7260	ANSHIKA KANOJIYA
64	UG-Computer Science and Engineering	BJ7268	AMIT KUMAR SINHA
65	UG-Computer Science and Engineering	BK5150	LAXMI
66	UG-Computer Science and Engineering	BJ7618	YOGESH KUMAR SONWANI
67	UG-Computer Science and Engineering	BH3470	SRISHTI DEWANGAN



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CSPTCL

CIN-U40108CT2003SGC015820, GST NO.22AADCC5773E1ZX

CHHATTISGARH STATE POWER TRANSMISSION COMPANY LIMITED.

(A GOVERNMENT OF CHHATTISGARH UNDERTAKING)

O/o EE 400 KV SUBSTATION Dn.KHEDAMARA, BHILAI

ADDRESS :400 KV S/S KHEDAMARA, JCW BHILAI-490024

Website-www.csptcl.co.in, E-Mail-ee400kvkhedamara@csptcl.co.in, ee400kvss.bhilai@gmail.com Phone-no.0788-2281553

NO.10-74/TRAINING/382

Bhilai, Date.25/08/2023

CERTIFICATE

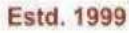
THIS IS TO CERTIFY THAT **Mr. CHANDAN SONI** STUDENT OF ELECTRICAL BRANCH 7TH SEMESTER OF **CHHATRAPATI SHIVAJI INSTITUTE OF TECHNOLOGY, DURG** HAS SUCCESSFULLY COMPLETED HIS/HER VOCATIONAL TRAINING DURING THE PERIOD FROM **19.07.2023 TO 17.08.2023** AT **400KV S/S DN. CHHATTISGARH STATE POWER TRANSMISSION COMPANY LIMITED KHEDAMARA BHILAI** IN ACCORDANCE TO ORDER OF DGM (HR) CSPTCL, RAIPUR ORDER NO.01-01/ORDER/AM-II/3735 RAIPUR DATED 18.07.2023.

WE WISH HIM/HER GREAT SUCCESS IN PROFESSIONAL JOURNEY.


(ER. HARISH KUMAR DEWANGAN)
EXECUTIVE ENGINEER
400 KV SUB STATION DIVISION
CSPTCL KHEDAMARA BHILAI

SANTOSH
KUMAR
SHARMA

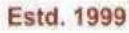
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LearnVern

Franchise of TOPS Technologies

CERTIFICATE OF TRAINING

This is to certify that

Ishwar Sahu

having CAN_2834040 has successfully completed training in

SolidWorks Course

from the period **Jul 2023 - Aug 2023**
at **LEARNVERN PVT LTD.**

Date of Issue: 11-08-2023

Certificate Id: CAN_28340406112510

Place: 6th Floor, Arjun Complex,
Shivranjani Cross Road,
Ahmedabad, Gujarat 380009



<https://www.learnvern.com/>

Signing Authority:



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Franchise of TOPS Technologies Pvt Ltd.
An Approved Training Partner of NSDC

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CERTIFICATE OF COMPLETION

Itronix Solutions certifies that

Aasik Raja Khan

Has successfully passed the requirements for

**AutoCAD 3D
Professional Certification**
offered by Itronix Solutions

Karan Arora

Founder & CEO



Certification Date **08/27/2023**

Certification ID **IS/CAD/23/187
27**



CERTIFIED
Digital Marketing
Associate

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SHARMA**

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DEPARTMENT OF CIVIL ENGINEERING INTERNSHIP SUMMARY

S.NO.	BATCH	NO. OF STUDENT
1	2020-24	1

S.NO.	NAME OF STUDENT	BATCH	TRAINING PERIOD	ORGANIZATION
1	RAHUL DEWANGAN	2020 - 2024	07.07.2023 – 05.08.2023	SECR

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इंजीनियरिंग
ENGINEERING

दक्षिण पूर्व मध्य रेलवे, रायपुर (छ.ग.) - 492008
SOUTH EAST CENTRAL RAILWAY, RAIPUR (C.G.) - 492 008

कार्यालय : वरिष्ठ मंडल इंजी. (सम.) / रायपुर
Office of the : Sr. DEN (Co-ord)/Raipur

No.SECR/R/Engg/General/1GS/313/Internship Civil Engg/



आज़ादी का
अमृत महोत्सव

Dt.26.03.2024.

TO WHOM SO EVER IT MAY CONCERN

This is to certify that **Shri Rahul Dewangan** [Roll No.300302020015] student of B.Tech. (Civil) 6th Sem. in Chhatrapati Shivaji Institute of Technology, Durg (C.G.) has attended **Field Training from 07.07.2023 to 05.08.2023** in South East Central Railway, Raipur Division in under mentioned locations:

Sl No.	Name of Work
1.	Constructions of Supela RUB No.442 in between Km.859/17-19 at Howrah-Mumbai Main Line.

Internship training of above named has been completed successfully.



(Aditya Kumar Tripathi)
Sr.Divl. Engineer(I)/Raipur
S.E.C.Railway pur

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CHHATRAPATI SHIVAJI INSTITUTE OF TECHNOLOGY, DURG DEPARTMENT OF CIVIL ENGINEERING INDUSTRIAL SITE VISIT

S.No.	NAME OF SITE	DATE
1	UTOPIA CONSTRUCTION, PULGAON CHOWK DURG	19.10.2024
2	RDC CONCRETE PVT. LMT. ,SILODA,RASMADA	14.05.2024
3	RMC CONSTRUCTION SITE KOHKA BHILAI	21.11.2022
4	RISHABH LUNIYA CONSTRUCTION SITE DURG	7.9.2021
5	PARKSYDE KACHANA, RAIPUR	12.9.2019
6	PWD CONSTRUCTION SITE, DHANORA DURG	16.03.2019
7	KADAM CAPITAL ANJORA, RAJNANDGAON	1.9.2018

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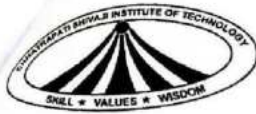
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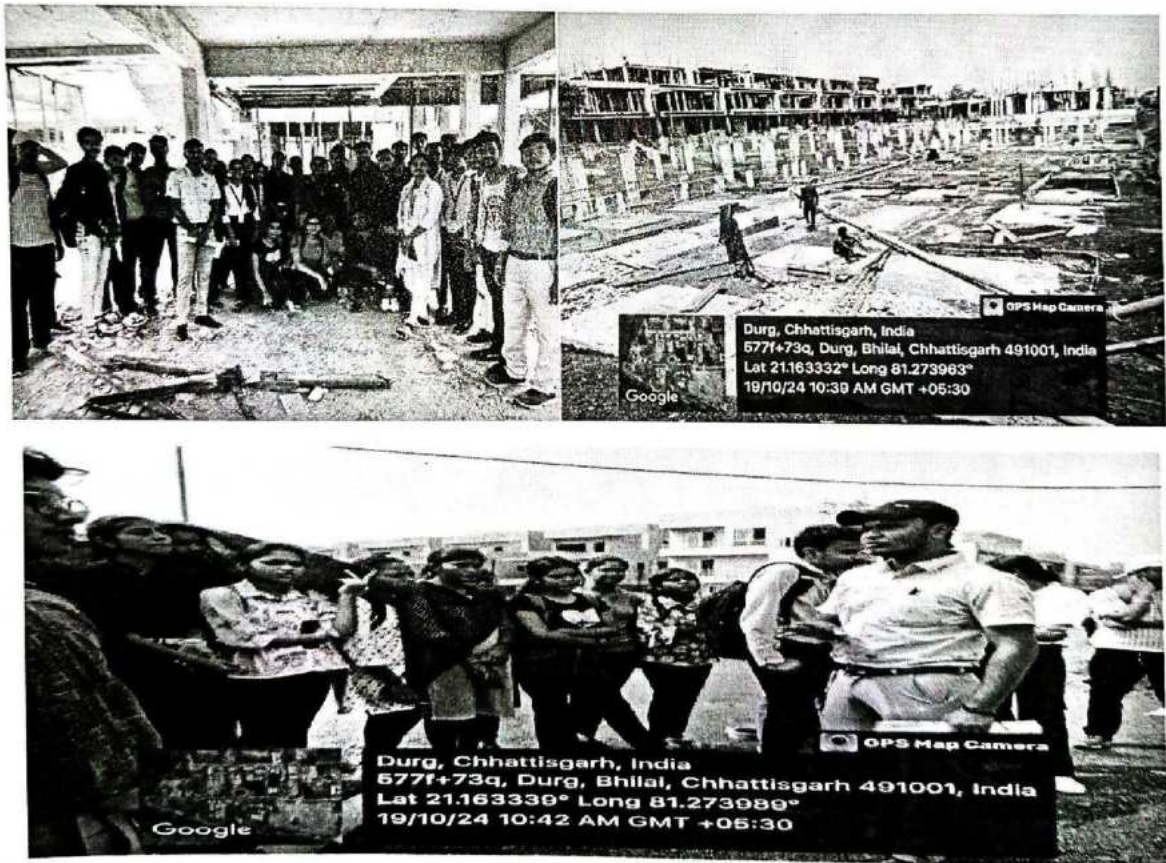
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The enthusiasm shown by the students during the visit is a testament to the effectiveness of such practical learning experiences. As future engineers and architects, they left the site with a deeper appreciation for the complexities and innovations in construction, better preparing them for their careers in the field.

Acknowledgments

We extend our gratitude to the management and staff of Utopia Durg for their hospitality and willingness to share their knowledge. Special thanks to the faculty members who guided the students throughout the visit, ensuring a rich learning experience.



Faculty Of Civil Engineering Department Explain the Technical Aspect of Site

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Mr. Pawan Kumar, Incharge Engg., Utopia Durg, Addressing the Students

HOD Civil Engg.

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DEPARTMENT OF CIVIL ENGINEERING

SITE VISIT REPORT

A site visit is conducted by the civil dept. for 4th and 6th semester students on the date of **14th May 2024 (Tuesday)** on **RDC CONCRETE PVT LTD.** Rasmada to improve the practical knowledge about preparation of concrete mix and the additives to be used for improving the strength of concrete.

RDC is a batch mixing plant in which ready mix concrete is prepared and directly dispatch to the site for the application. The visit is not only beneficial for civil engg. Students but also for mechanical point of view. As the entire mechanism of the plant is based on mechanical Engg. As well.

The students are guided by the senior technical officer Mr. Shyamanand Ghosh and faculty members, where the students observed

1. The students are first taken to the conference hall where they all brief about the national cube test competition, and also give the student hint regarding the preparation of mixture and its proportion.
2. The students are directly taken to the plant where the mixing process is done and make aware of all the parts of the machinery and its utility.
3. Later on the students were told about admixtures and additives with its importance.
4. After the site visit all the students were taken into their control room to make them aware how the mixing done and loaded in the mixer.
5. After that students were visited to there labs where they make familiar about the modern technologies and machines used for the strength testing. Such as CTM machine (automatic), vibrator (semi automatic) etc . after this the visit comes to an end with group photo.

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Lat 21.189482°

Long 81.219438°

14/05/24 12:05 PM GMT +05:30

GPS Map Camera

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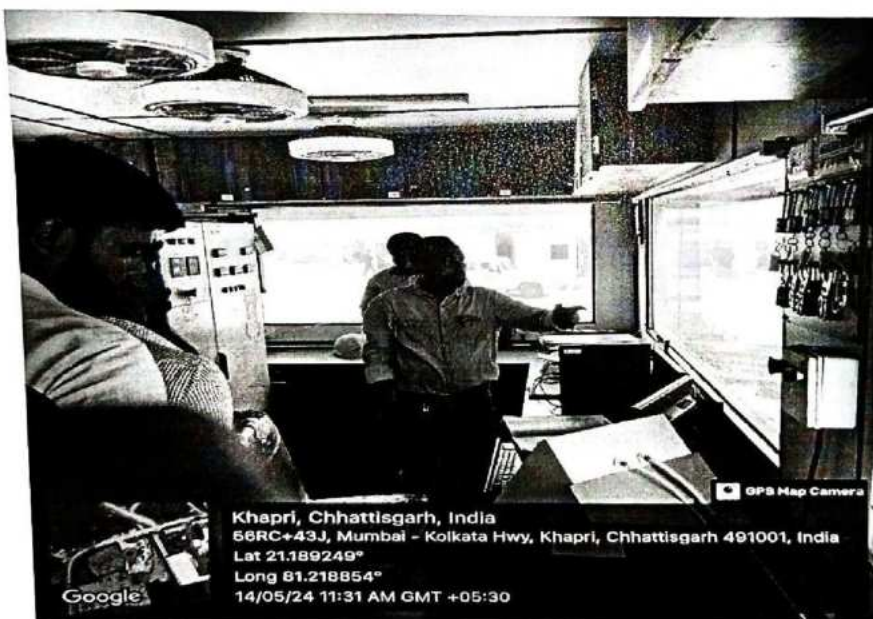
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56RC+43J, Mumbai - Kolkata Hwy, Khapri, Chhattisgarh 491001, India
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Long 81.218854°
14/05/24 11:31 AM GMT +05:30

Students are guided by the Site Engineer and Faculty Members


HOD CIVIL

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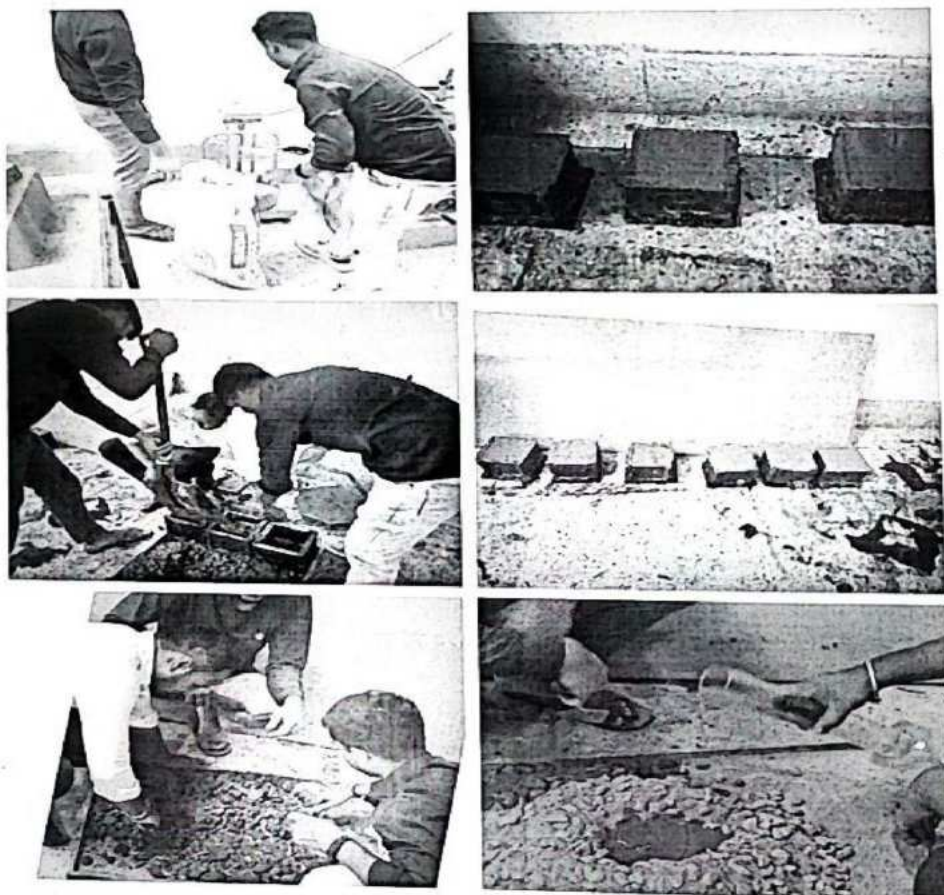
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Making Process of Concrete and Cube Casting

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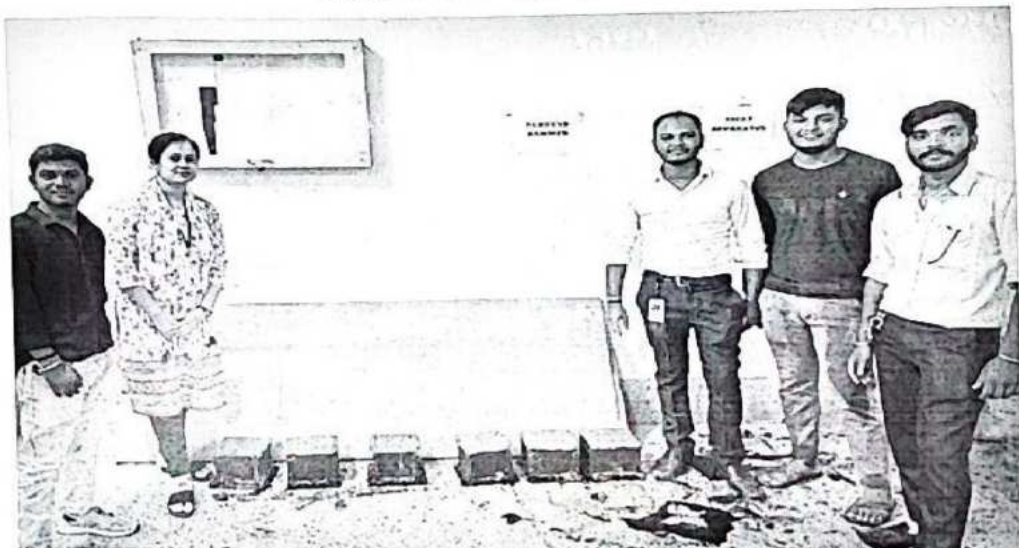
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Cube Casting by Students, under the Guidance of Faculties of Civil Engg. Dept.

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Understanding Earthquakes: An Add-On Course

Chhattisgarh Swami Vivekananda Technical University, Bhilai

Branch: Common to all Branches

Semester: B.Tech 4th Sem

Subject: ENGINEERING GEOLOGY

Code: B0205415(020)

Understanding Earthquakes: An Add-On Course

Course Overview:

This add-on course aims to provide a deeper understanding of earthquakes, their causes, effects, and mitigation strategies. It will delve into the geological processes behind earthquakes, their impact on society and infrastructure, and ways to minimize their consequences.

Course Duration:

- Duration: 4 weeks
- Frequency: Weekly sessions

Course Content:

Week 1: Introduction to Earthquakes

- Understanding the Earth's structure and tectonic plates
- Types of seismic waves and their characteristics
- Historical earthquakes and their significance

Week 2: Causes and Mechanics of Earthquakes

- Plate tectonics and fault lines
- Different types of faults: normal, reverse, and strike-slip
- Stress accumulation and release: the earthquake cycle

CLASS-1

Introduction to Earthquakes

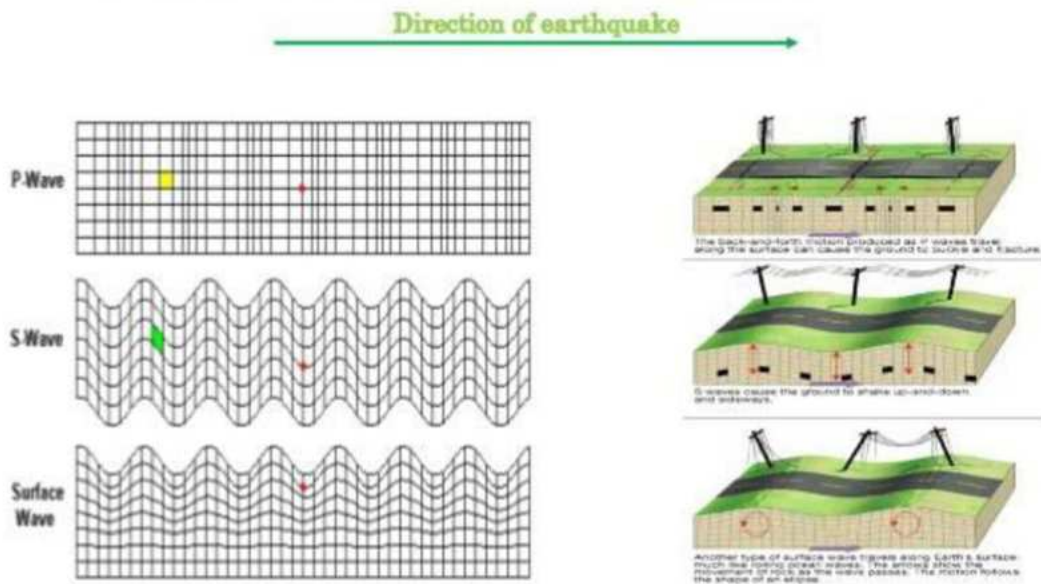
Understanding the Earth's structure and tectonic plates:

Understanding earthquakes begins with understanding the Earth's structure. The Earth is composed of several layers, including the solid inner core, liquid outer core, mantle, and crust. The crust is divided into tectonic plates, which float on the semi-fluid asthenosphere beneath them. These plates constantly move and interact with each other, leading to geological phenomena such as earthquakes, volcanic eruptions, and mountain formation.

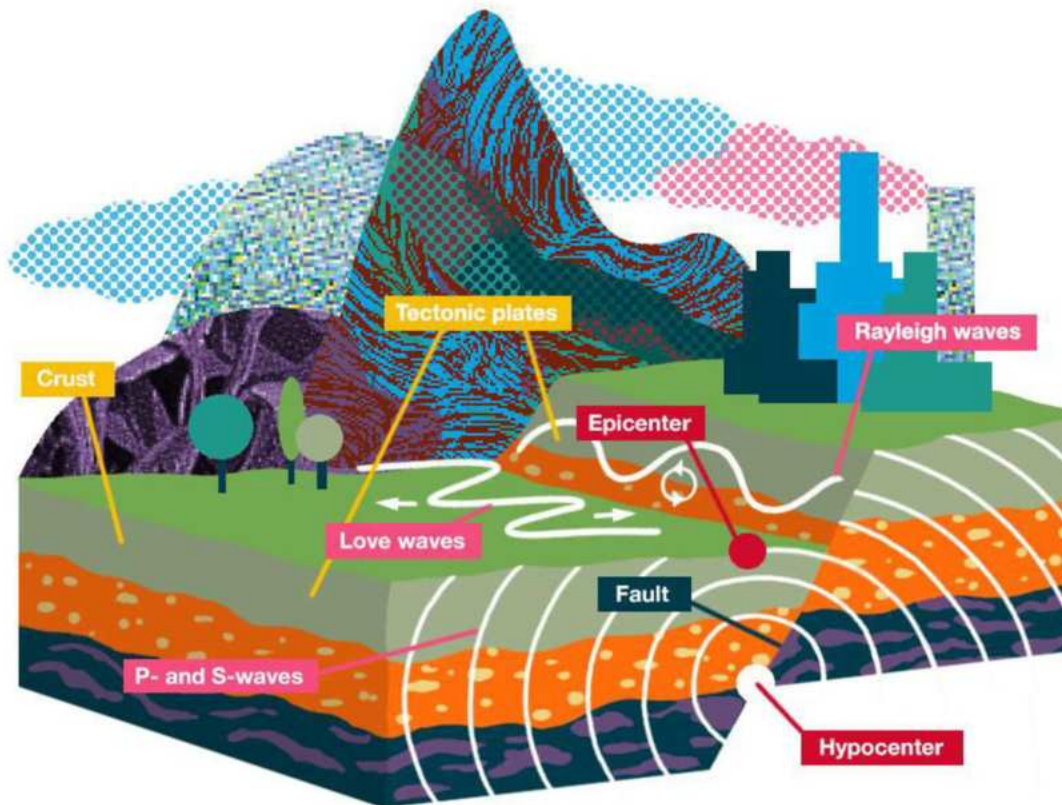
Types of seismic waves and their characteristics:

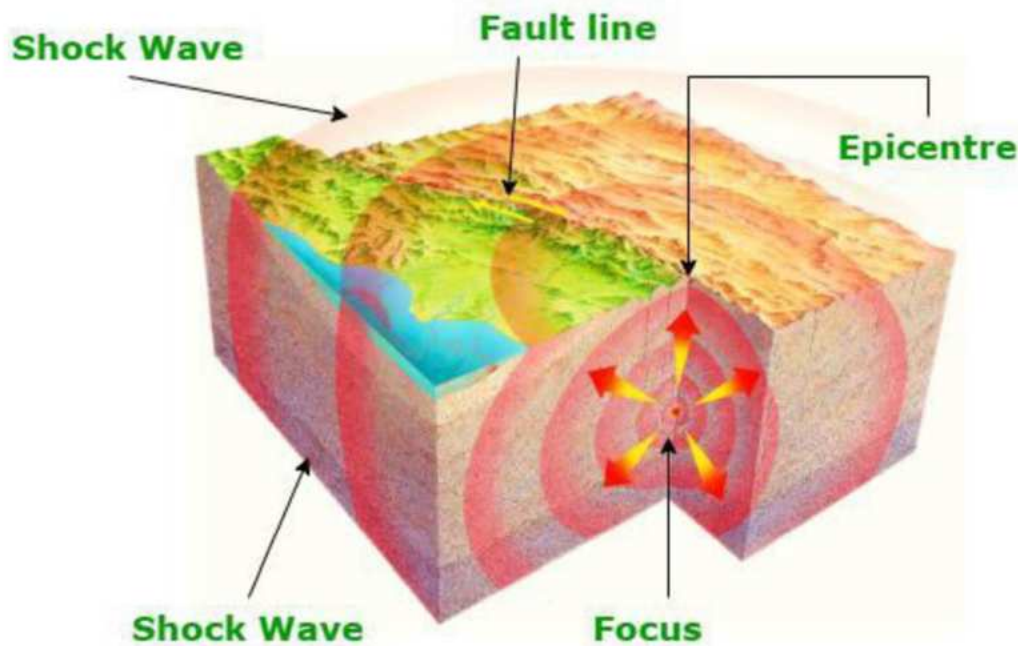
Seismic waves are the vibrations that travel through the Earth's interior during an earthquake. There are two main types: body waves and surface waves. Body waves include primary (P-waves) and secondary (S-waves), which travel through the Earth's interior. P-waves are compressional waves, while S-waves are shear waves. Surface waves, which travel along the Earth's surface, include Love waves and Rayleigh waves. Each type of wave has distinct characteristics, including speed, amplitude, and direction of motion, which scientists use to study and understand earthquakes.

3 types of seismic waves



critical point, known as the elastic limit. Beyond this point, the rocks rupture suddenly, releasing stored energy in the form of seismic waves, resulting in an earthquake. Following an earthquake, the cycle begins anew, with stress accumulating along the fault line until it reaches another breaking point. Understanding the earthquake cycle is crucial for assessing seismic hazards, predicting future earthquakes, and implementing effective mitigation strategies to minimize their impact on society.





CLASS-3

Ground shaking, liquefaction, and landslides:

Ground shaking is one of the most immediate and noticeable effects of an earthquake. As seismic waves travel through the Earth, they cause the ground to shake, resulting in structural damage to buildings and infrastructure. Liquefaction is a phenomenon where saturated soil temporarily loses strength and behaves like a liquid due to the intense shaking during an earthquake, leading to ground instability and potential building collapse. Landslides can also occur as a result of ground shaking, especially in areas with steep slopes or unstable terrain, posing additional hazards to communities.

Tsunamis: formation, characteristics, and mitigation:

Tsunamis are large ocean waves generated by undersea earthquakes, volcanic eruptions, or underwater landslides. When an earthquake occurs beneath the ocean floor, it displaces water, creating a series of waves that can travel across vast distances at high speeds. Tsunamis can cause devastating coastal flooding and widespread destruction when they reach shorelines. Mitigation measures for tsunamis include early warning systems, coastal land-use planning, and public education to ensure communities are prepared to evacuate to higher ground in the event of a tsunami warning.

Impact on buildings, infrastructure, and human lives:

Earthquakes can have significant impacts on buildings, infrastructure, and human lives, depending on their magnitude, depth, and proximity to populated areas. Buildings may suffer structural damage or collapse, leading to injuries and fatalities among occupants. Infrastructure such as roads, bridges, and utilities can also be severely affected, disrupting transportation networks and essential services. The socio-economic impact of earthquakes can be profound, causing displacement, loss of livelihoods, and long-term recovery challenges for affected communities. Adequate building codes, retrofitting of existing structures, and disaster preparedness initiatives are essential for mitigating the impact of earthquakes and ensuring the resilience of communities in earthquake-prone regions.





CLASS-4

Earthquake Preparedness and Mitigation

Seismic risk assessment and hazard mapping:

Seismic risk assessment involves evaluating the likelihood and potential consequences of earthquakes in a particular region. This process includes analyzing historical seismic activity, geological features, and local infrastructure vulnerabilities. Hazard mapping identifies areas prone to ground shaking, liquefaction, landslides, and tsunamis, providing crucial information for land-use planning, emergency management, and public awareness campaigns. By understanding seismic hazards, communities can implement targeted mitigation strategies to reduce risks and enhance resilience to earthquakes.

Building codes and retrofitting techniques:

Building codes are regulations that specify minimum standards for the design, construction, and maintenance of structures to withstand seismic forces. These codes incorporate seismic-resistant design principles, such as reinforced concrete, flexible building materials, and appropriate foundation systems, to minimize damage and ensure occupant safety during earthquakes. Retrofitting involves strengthening existing buildings to meet current seismic standards, often through structural modifications, foundation upgrades, or installation of seismic dampers and bracing systems. Implementing and enforcing rigorous building codes and retrofitting

Certificate Course on "COURSE ON HTML,CSS,PHP BY S2P EDU TECH ,NAGPUR"

Year	Certificate	Schedule	Duration	Organization	Venue	Resource Person
B.Tech, 3 rd ,4 th year, CSE,IT	Course on "COURSE ON HTML,CSS,PHP BY S2P EDU TECH , NAGPUR"	11 sep. 2020 to 19 Sep. 2020	40Hours	CSIT,Durg	Lab No. 8	Mr Gandhar Patwardhan

Course Coordinator

HOD
Department of computer Science
CSIT Durg

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Run By: Shivnath Shikshan & Seva Samiti

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11th September 2020 to 19th September 2020

For Final & Pre-Final Year Students [CSE & IT Department]

RESOURCE PERSON:



Mr. Gandhar Patwardhan
Founder, Managing Director, Trainer
at S2P EDUTECH

CO-ORDINATOR



Mr. Kranti Kumar Jain
Assistant Professor
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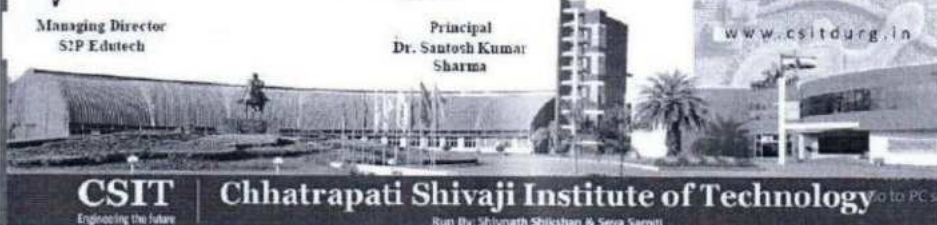
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Certificate of Participation

This is to Certify that Mr. / Ms. **MUSKAN RATHI** of **Information Technology Department** of Year **3rd** has participated in the "**COURSE ON HTML, CSS, PHP BY S2P EDU TECH**" from **11/09/2020 to 19/09/2020** Organised by Chhatrapati Shivaji Institute of Technology, Durg.


Managing Director
S2P Edutech


Principal
Dr. Santosh Kumar Sharma



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CERTIFICATE COURSE ON “ENERGY AUDIT”

Year	Certificate Course	Schedule	Duration	Organization	Venue	Resource Person
B. Tech. 2 nd & 3 rd year	COURSE ON “ENERGY AUDIT”	4/02/2019 to 14/02/2019	40 Hours	CSIT,DURG	Seminar Hall (G2)	Mr. AshishBafna


Course Coordinator


HOD
Electrical Engineering Department
Head of Department
Electrical & Electronics Engg
CSIT, DURG

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SPEAKER

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Co-Ordinator
Sanjay Kumar Singh

Director
Dr. Santosh Kumar Sharma



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**04th to 14th February
2019**

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Dr. Sangeeta Kaul
Director



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Vasant Kunj, New Delhi 110070, India

Tel: 91-11-26742222, 26741266

91-9810329992 (Mobile)

E-mail: sangs@delnet.ren.nic.in,

sangskaul2003@yahoo.co.in

Web: www.delnat.in

March 3, 2023

DELNET/IM-6766/cgCSII/ME/M/2023

Sub: DELNET Membership Renewal

Dear Mr. Madharia,

We acknowledge with thanks the receipt of ₹ 13,570 (₹ Thirteen Thousand Five Hundred Seventy only) received through NEFT dated 3.3.2023 made towards the DELNET Annual Institutional Membership Fee for the period 6.3.2023 to 5.3.2024. The receipt no. 73510 dated 3.3.2023 is enclosed for the office records.

You are requested to access DELNET databases through the World Wide Web using the following procedure:

Web Address: <http://www.delnat.in>

Click onto "New Discovery Portal". Since the IP address provided by you is not static (broadband), you are requested to use following login & password to access the new discovery portal of DELNET.

Login : cgcsit
Password : csit6766

Kindly note your Inter Library Loan (ILL for Books) Password is "cgcsitlib" to be used while registering a request. You are also welcome to send us the bibliographic references at sangs@delnet.ren.nic.in, sangskaul2003@yahoo.co.in for the resources needed by you. We will try our best to locate these resources. We would like to inform you that DELNET has launched WELBIVW Youtube channel which contains a large number of video recordings of Webinars organised by DELNET including sessions on DELNET resources and services. The link is available at the Discovery Portal. We would further like to inform you that Usage Report can be generated through "USAGE STATISTICS" link which appears at the top side of the landing page of the discovery portal. Kindly use the password as 6766***1992 to download the pdf, containing usage report of your institution.

I would like to mention that DELNET provides access to more than three crore catalogue records of books, journals, articles, etc. through Discovery Portal and also more than one crore and fifty lakh full-text e-books, e-journals & e-articles through Knowledge Gainer Portal. We are pleased to inform you that DELNET has also recently launched ViSiOn Portal which contains Video Recordings of Lectures on varied subjects. We request you to kindly contribute the video lectures of the Faculty Members of your Institution. Kindly contact us for further assistance. DELNET also provides Delplus software free of charge for library automation purpose. DELNET Guest House facility at New Delhi can also be availed by member-libraries on payment basis.

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With kind regards,

Yours sincerely,

Sangeeta Kaul

Mr. Anshu Nandan Madharia
Librarian

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March 3, 2023

DELNET 30 Years

Sangeeta Kaul
Director

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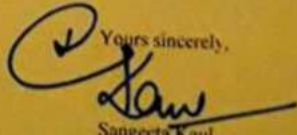
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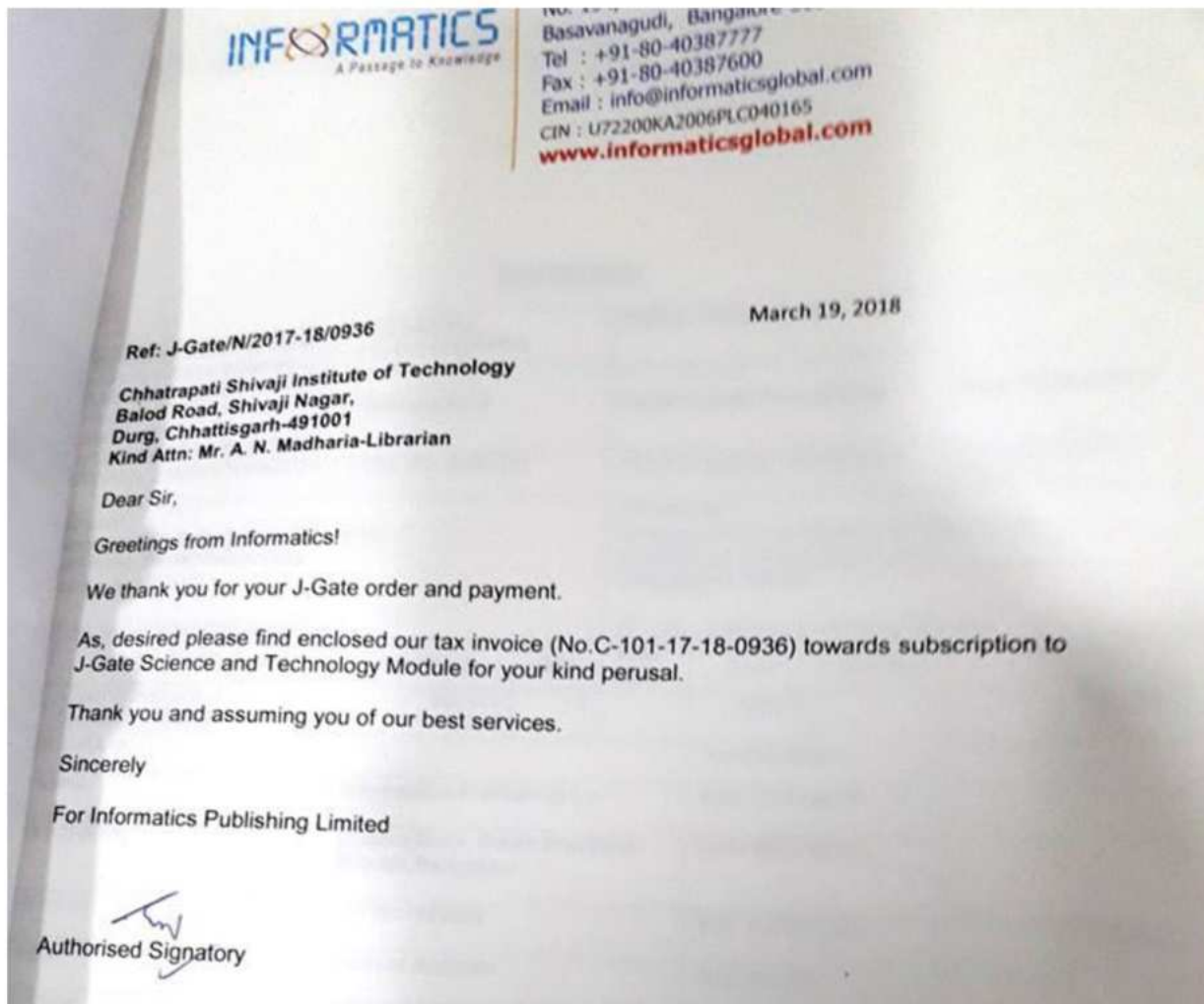
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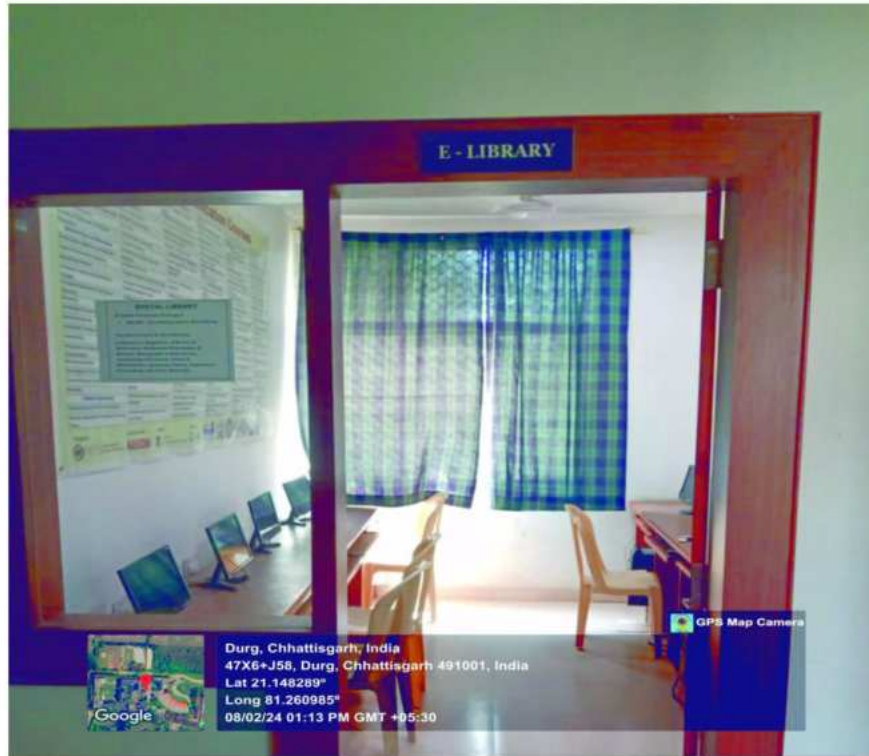
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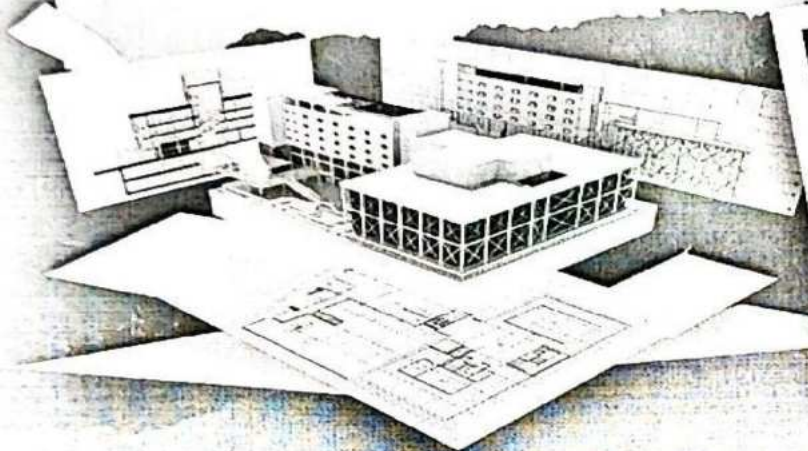


Chhatrapati Shivaji Institute of Technology



WORKSHOP

Resource Person



Mr. PRAVEEN MAHAPATRA
Assistant Professor
Lakhmi Chand Institute of Technology, Bilaspur



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Mr. Praveen Mahapatra addressing the students of 3rd & 4th year on AutoCAD & SKETCHUP software at Lab 04 CSIT Durg



Mr. Praveen Mahapatra addressing the students of 3rd & 4th year on AutoCAD & SKETCHUP software at Lab 04 CSIT Durg

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Certificate distribution of 3rd Year Students



Certificate distribution of 4th Year Students

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A Report on Guest Lecture

Held On: 23/11/2023

Name of Guest: Dr. S. Gangopadhya

Designation: Assistant Engineer CREDA Raipur

Topic: "PVD COATINGS FOR CUTTING TOOLS, MICROMACHINING, SURFACE INTEGRITY
IN MANUFACTURING"

Venue: Sambhaji Hall

Time: 03:00 PM

The scenario of the guest lecture was started by honoring the respected guest Dr. S. Gangopadhya, our respected Dean of research Mr. H R Chandrakar & HOD Mr. Pradeep Agrawal, along with the respected faculties of our department. The program was proceeded by the speech of our respected Mr. H R Chandrakar Sir. Then before the starting of the guest lecture a brief introduction of honorable guest was given by Ms. Vasudha Verma,. After this, honorable Dr. S. Gangopadhya, proceeded the further program & shared his experience & knowledge with the students. He firstly gave the brief introduction about Physical Vapor Deposition (PVD) coatings are widely used in the manufacturing industry to enhance the performance and lifespan of cutting tools. These coatings provide numerous benefits, including increased hardness, wear resistance, and reduced friction. This report will explore the application of PVD coatings in cutting tools, their impact on micromachining, and their role in maintaining surface integrity during manufacturing processes. Then he cleared the doubt of the students which were arise during the lecture. Then vote of thanks was given by our respected HOD Mr. Pradeep Agrawal.

After this, Dr. S. Gangopadhya shown the entire students PVD Coatings for Cutting Tools, so that he could give the practical information & could clear the doubt of students. The session came to end at about 4:45 PM.

COORDINATOR

Abhishek K. Gupta.

HOD

MECH/MEX Dept.

(Mr. Pradeep Agrawal)

Head of the Department

Mechanical Engineering

C S I T, Durg

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LIST OF STUDENTS UNDERGONE PROJECT WORK FOR THE ACADEMIC YEAR 2022-23

S.No.	Programme name	Program Code	List of students undertaking project work
1	B.TECH. ELECTRICAL and ELECTRONICS ENGINEERING	BT3025	AJEET VERMA
2	B.TECH. ELECTRICAL and ELECTRONICS ENGINEERING	BT3025	MANEESH KEWAT
3	B.TECH. ELECTRICAL and ELECTRONICS ENGINEERING	BT3025	SAGAR JAISWAL
4	B.TECH. ELECTRICAL and ELECTRONICS ENGINEERING	BT3025	VINAY KUMAR
5	B.TECH. ELECTRICAL and ELECTRONICS ENGINEERING	BT3025	CHANDRIKA
6	B.TECH. ELECTRICAL and ELECTRONICS ENGINEERING	BT3025	DEEPIKA
7	B.TECH. ELECTRICAL and ELECTRONICS ENGINEERING	BT3025	MAITREE
8	B.TECH. ELECTRICAL and ELECTRONICS ENGINEERING	BT3025	DHARMENDRA KUMAR
9	B.TECH. ELECTRICAL and ELECTRONICS ENGINEERING	BT3025	JAI DESHMUKH
10	B.TECH. ELECTRICAL and ELECTRONICS ENGINEERING	BT3025	SURYAKANT TAMRAKAR
11	B.TECH. ELECTRICAL and ELECTRONICS ENGINEERING	BT3025	P.YAMINI
12	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	GAGAN SHUKTEL
13	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	ABHISHEK RAI
14	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	AYUSH YADAV
15	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	D KRISHANU PRABHU
16	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	BHOOMIKA SAHU
17	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	RAKSHA SHARMA
18	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	RISHABH SHRIVASTAVA
19	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	YOGESH CHANDRAKAR
20	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	ABDUL RAHMAN
21	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	KEHKASA ANJUM
22	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	HARSH KUMAR DESHMUKH
23	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	KOSTA AADHIPRITHVISRI DEWANGAN
24	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	VINI NAYAK
25	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	DHALENDRA KUMAR MESHRAM
26	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	VIDHI SAWAI
27	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	SANJANA GOVINDANI
28	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	PRITI KUNDU
29	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	HEMLATA VERMA
30	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	SHRIYA THAKUR

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S.No.	Programme name	Program Code	List of students undertaking project work
31	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	R TUSHAR
32	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	PREYASH KUMAR
33	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	AMAN SAHOO
34	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	LUCKY SAHU
35	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	AFROZ ANSARI
36	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	AYUSH CHANDRA
37	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	ADITI YADAV
38	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	AKASH SHRIVASTAV
39	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	Abhishek Singh
40	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	KAVITA Sahu
41	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	HITESH Verma
42	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	AKASH AGRAWAL
43	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	VEENA SAHU
44	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	GUNJAN VERMA
45	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	SHUBHI SINGH
46	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	ANCHAL SINGH BAGHEL
47	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	AMRITA PANDEY
48	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	AAKANSHA LAL NETAM
49	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	NAMAN KHANDLWAL
50	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	GITESH KUMAR
51	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	VEDANK PACHKAWADE
52	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	TUSHAR YADAV
53	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	ANKIT NAYAK
54	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	SURAJ VERMA
55	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	PRAVEEN VERMA
56	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	ROHIT SHRIVASTAVA
57	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	NIKHIL LILWANI
58	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	VAIBHAV SULE
59	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	P HARSHIT KUMAR
60	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	MVS VARUN
61	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	ASHTIK KUMAR PRAJAPATI
62	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	CHANDAN DHURV
63	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	DAYANIDHI SAHU
64	B.TECH. COMPUTER SCIENCE AND ENGINEERING	BT3022	ANKIT KUMAR
65	B.Tech. Mechatronics Engineering	BT067	Jayant Sahu

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S.No.	Programme name	Program Code	List of students undertaking project work
66	B.Tech. Mechatronics Engineering	BT068	Kuldeep Sahu
67	B.Tech. Mechatronics Engineering	BT069	Dristy Pal
68	B.Tech. Mechatronics Engineering	BT070	Eshita Godsore
69	B.Tech. Mechatronics Engineering	BT071	Sannidhi Pushpay
70	B.Tech. Mechanical Engineering	BT3037	KAUSHAL DEWANGAN
71	B.Tech. Mechanical Engineering	BT3038	KUNAL DEWANGAN
72	B.Tech. Mechanical Engineering	BT3039	UTKARSH DESHMUKH
73	B.Tech. Mechanical Engineering	BT3040	DEVENDRA KUMAR VERMA
74	B.Tech. Mechanical Engineering	BT3041	PRANJAL PANDEY
75	B.Tech. Mechanical Engineering	BT3042	PRANESH KUMAR SAHU
76	B.Tech. Mechanical Engineering	BT3043	L DEEPAK NARAYANA
77	B.Tech. Mechanical Engineering	BT3044	ROHIT KUMAR SAHU
78	B.Tech. Mechanical Engineering	BT3045	VARSHA RANI SAHU
79	B.Tech. Mechanical Engineering	BT3046	DALLIYCHAND
80	B.Tech. Mechanical Engineering	BT3047	DUPESH KUMAR DEWANGAN
81	B.Tech. Mechanical Engineering	BT3048	RAHUL KUMAR
82	B.Tech. Mechanical Engineering	BT3049	YAGYA KUMAR
83	B.Tech. Mechanical Engineering	BT3050	PIYUSH VERMA
84	B.Tech. Mechanical Engineering	BT3051	HULENDRA
85	B.Tech. Mechanical Engineering	BT3052	NITISH KUMAR SAHU
86	B.Tech. Mechanical Engineering	BT3053	VIRENDRA PANDIT
87	B.Tech. Mechanical Engineering	BT3054	MANISH SAHU
88	B.Tech. Mechanical Engineering	BT3055	VINAY SINGH
89	B.TECH. CIVIL ENGINEERING	BT3020	Dharmendra Chandrakar
90	B.TECH. CIVIL ENGINEERING	BT3020	Yashpal Gangeshwar
91	B.TECH. CIVIL ENGINEERING	BT3020	Yugal Kishor Sahu
92	B.TECH. CIVIL ENGINEERING	BT3020	Jafar khan
93	B.TECH. CIVIL ENGINEERING	BT3020	Bhisham Ratre
94	B.TECH. CIVIL ENGINEERING	BT3020	Md. Hasnain Ashfaq
95	B.TECH. CIVIL ENGINEERING	BT3020	Mayank Soni
96	B.TECH. CIVIL ENGINEERING	BT3020	Rishabh Manikpuri
97	B.TECH. CIVIL ENGINEERING	BT3020	Saurabh Balekar
98	B.TECH. CIVIL ENGINEERING	BT3020	Rishabh Deshmukh
99	B.TECH. CIVIL ENGINEERING	BT3020	Khilendra verma
100	B.TECH. CIVIL ENGINEERING	BT3020	Avinash Markam

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S.No.	Programme name	Program Code	List of students undertaking project work
101	B.TECH. CIVIL ENGINEERING	BT3020	Shweta Dongre
102	B.TECH. IT ENGINEERING	BT3033	YASHWANT KUMAR BELGAHE
103	B.TECH. IT ENGINEERING	BT3034	RAHUL KUMAR KAUSHIK
104	B.TECH. IT ENGINEERING	BT3035	DANI RAM SAHU
105	B.TECH. IT ENGINEERING	BT3036	HOSHANG KUMAR SAHU
106	B.TECH. IT ENGINEERING	BT3037	SANKALP SHARMA
107	B.TECH. IT ENGINEERING	BT3038	MONIKA VERMA
108	B.TECH. IT ENGINEERING	BT3039	AYESHA PARVEEN MIRZA
109	B.TECH. IT ENGINEERING	BT3040	MADHU
110	B.TECH. IT ENGINEERING	BT3041	DEEPIKA SAHU [COB] EEE TO IT
111	B.TECH. IT ENGINEERING	BT3042	KHILESH SAHU
112	B.TECH. IT ENGINEERING	BT3043	PURUSHOTTAM DHIVER
113	B.TECH. IT ENGINEERING	BT3044	AVINASH SAHU
114	B.TECH. IT ENGINEERING	BT3045	SHRADDHANJALI
115	B.TECH. IT ENGINEERING	BT3046	VIGYA RAIPURIYA
116	B.TECH. IT ENGINEERING	BT3047	ROHIT SONI
117	B.TECH. IT ENGINEERING	BT3048	DURGESH SAHU
118	B.TECH. IT ENGINEERING	BT3049	MAHIMA
119	B.TECH. IT ENGINEERING	BT3050	KARAN KUMAR SAHU
120	B.TECH. IT ENGINEERING	BT3051	VIVEK KUMAR YADAV

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A

Major Project Report

On

Fabrication and Analysis of Solar and Iot based Automatic Cattle Feeder

Submitted to

CHHATTISGARH SWAMI VIVEKANAND TECHNICAL UNIVERSITY

BHILAI (C.G.), INDIA

In partial fulfillment of requirement for the award of degree

of

Bachelor of Technology

in

Electrical & Electronics Engineering

By

Ajeet Verma

Maneesh Kewat

Sagar Jaiswal

Vinay Kumar

Under the Guidance of

Er. Ashraf Jafri

Dean-research & development

Associate Professor

Electrical & Electronics Engineering



Department of Electrical & Electronics Engineering

Chhatrapati Shivaji Institute of Technology, Durg

Shivaji Nagar, Balod Road, Durg (C.G.), 491001

Session 2022-2023

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CERTIFICATE

This is to certify that the report of the project submitted is an outcome of the project work entitled **Fabrication and Analysis of Solar and Iot Based Automatic Cattle Feeder** carried out by

Ajeet Verma
Maneesh Kewat
Sagar Jaiswal
Vinay Kumar

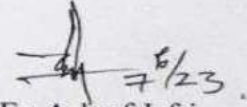
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Roll No. : 300302519011
Roll No. : 300302519015
Roll No. : 300302519018

Enrollment No.:BH4748
Enrollment No.:BH4758
Enrollment No.:BH4762
Enrollment No.:BH4765

Carried out under my guidance and supervision for the award of Degree in Bachelor of Technology in Electrical & Electronics of Chhattisgarh Swami Vivekanand Technical University, Bhilai (C.G.), India.

To the best of my knowledge the report

- Embodies the work of the candidate him/herself,
- Has duly been completed,
- Fulfills the requirement of the Ordinance relating to the BE degree of the University and
- Is up to the desired standard for the purpose of which is submitted.


Er. Ashraf Jafri

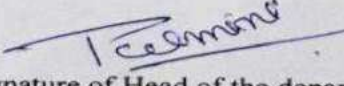
Dean-research & development

Associate Professor

Department of Electrical and Electronics.

Chhatrapati Shivaji Institute of Technology, DURG

The project work as mentioned above is hereby being recommended and forwarded for examination and evaluation.


(Signature of Head of the department with seal)
Head of Department
Electrical & Electronics Engg
CSIT, DURG

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CERTIFICATE BY THE EXAMINERS

This is to certify that the project work entitled

Fabrication and Analysis of Solar and IOT Based Automatic Cattle Feeder

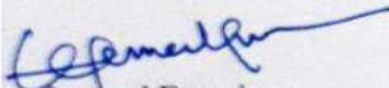
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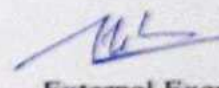
Ajeet Verma
Maneesh Kewat
Sagar Jaiswal
Vinay Kumar

Roll No. : 300302519001
Roll No. : 300302519011
Roll No. : 300302519015
Roll No. : 300302519018

Enrollment No.: BH4748
Enrollment No.: BH4758
Enrollment No.: BH4762
Enrollment No.: BH4765

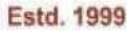
Has been examined by the undersigned as a part of the examination for the award of Bachelor of Technology degree in Electrical & Electronics Engineering of Chhattisgarh Swami Vivekanand Technical University, Bhilai.


Internal Examiner
Date: 8th June 2023


External Examiner
Date: 8/6/23

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MSME TECHNOLOGY CENTRE, DURG
सूक्ष्म, लघु एवं मध्यम उद्यम मंत्रालय, भारत सरकार की सोसायटी
Ministry of Micro Small and Medium Enterprises, Government of India Society



Ref. No. MSMETC(D)/Workshop/03-24/ 694

Date: 23.03.2024

To,

Mr. Sanjay Singh
Training & Placement Officer
Chhatrapati shivaji Institute of Technology, Durg

Sub: Proposal to conduct one day workshop on advanced technologies and Tools for the students of your institution.

Sir,

With reference to our telephonic conversation with you on dated 23.03.2024 regarding conduction of workshop on advanced technologies and Tools, we hereby propose to conduct two days workshop for the students of your esteemed College/Institution at your premises. This workshop aims to provide them with valuable insights as well as hands-on practice in the following areas.

Topics:

1. CNC Programming
2. 5S System (Workplace organization)
3. Geometrical Dimensioning & Tolerance
4. Tolerance Stackup
5. Design Failure Mode Effect & Analysis
6. 7 Quality Control Tools
7. Robotics
8. Solar PV
9. Programmable Logic Control (PLC)
10. Python Programming
11. Internet of Things (IoT)
12. CNC Programming

The workshop aims to achieve the following objectives:

- I. Introduce students to the latest trends and advancements in technology.

Address: -Plot No. 2D, Sector-B, Boral Industrial Growth Centre, Village-Rasmada, Dist.-Durg, Chhattisgarh, Pin-491001
Phone: 0788-2617200, Email: info@msmetcdurg.org Website: www.msmetcdurg.org

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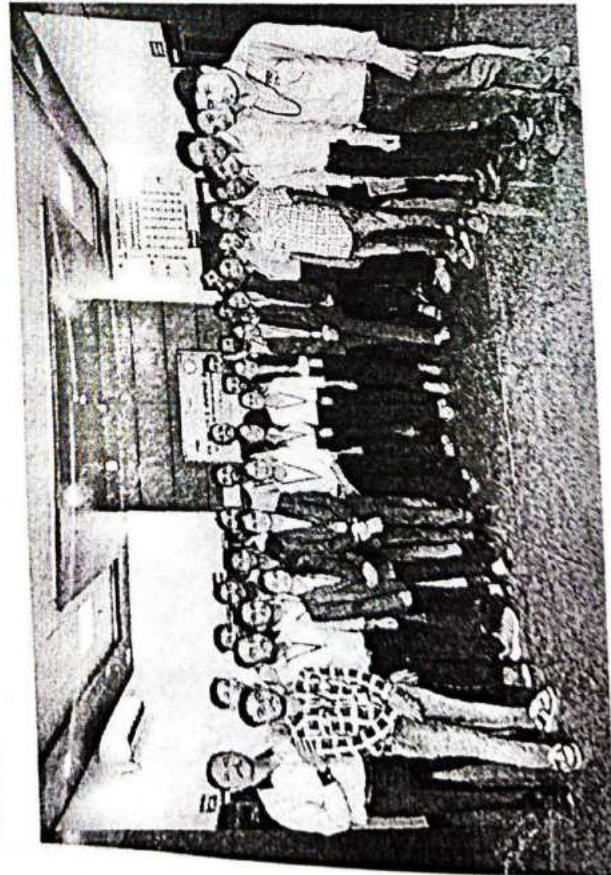
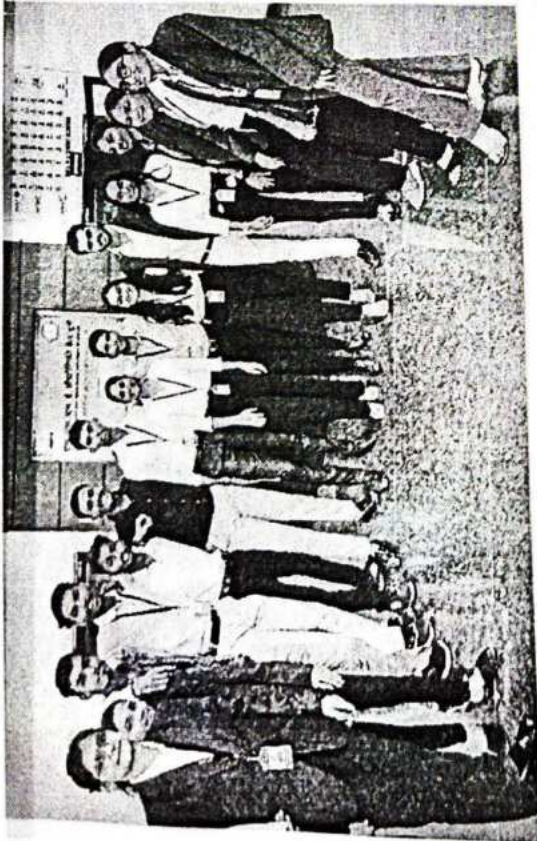
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Students at the Rasmada Unit of MSME.

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Serial No. 131488

Roll No. DWS132404001488

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Ministry of Micro, Small and Medium Enterprises,
Government of India

A Govt. of India Society

UNDER

Ministry Of Micro Small & Medium Enterprises,

Plot - 2D, Sector-B, Borai Industrial Growth Centre, Rasmada, Durg (C.G)-491001

Certificate of Participation

This is to certify that Mr./Ms. Ayush Kumar of CSIT

Durg has participated in "One day workshop / Seminar"

on 101 & Python held on 11/05/2024


Co-ordinator


Sr. Manager (Training)

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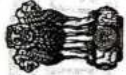
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Serial No 13.14.87...

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Certificate of Participation

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Nitin chakradhari of CSIT

Durg

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on TOT & Python

held on 11/5/2024.

Santosh Kumar Sharma
Co-ordinator

Sr. Manager (Training)

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CHHATRAPATI SHIVAJI INSTITUTE OF TECHNOLOGY, DURG
BRANCH: MECHANICAL ENGINEERING **SEMESTER: 7TH**
SUBJECT: MACHINE TOOL TECHNOLOGY
SUBJECT CODE: D037731 (037)

ASSIGNMENT NO 1

1. Explain different types of chips produced during turning. What is the use of chip-breaker
2. In an orthogonal cutting process, following data were observed-Chip length of 80 mm was obtained with an uncut chip length of 200 mm and the rake angle 20° and depth of cut 0.5 mm. The horizontal and vertical components of force F and F_y were 2000 N and 200 N respectively. Determine the shear angle, chip thickness, friction angle and resultant cutting force
3. Sketch a single point cutting tool and show various angles thereon and discuss their significance.
4. During metal cutting test under orthogonal cutting the following observations were made:

Cutting velocity 1100 N
Feed Force-1020 N
Rake angle $=10^\circ$
Shear plane angle 19°
Cutting velocity 165 meter per minute
Velocity of shear $=164.5$ meter per minute
Velocity of chip-54.4

Show that the work input is equal to the sum of work done in shearing and work done against friction.

5. What is chip formation and explain different types of chips produced during turning? What is the use of chip-breaker?

Ans of unit-01

Assignment No-01

VED PRAKASH

Roll no: 300303721001

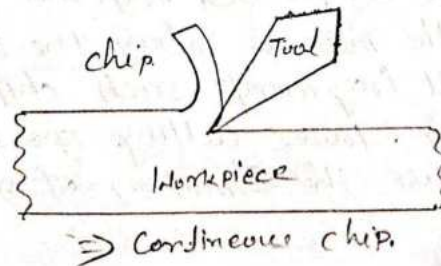
Ans-b

Ans:- Chip formation:- the type of chip produced during metal cutting depends upon the machining conditions & material being cut. the variable which influence the type of chip produced are as follow:-

1. properties of material cut especially ductility.
2. Depth of cut.
3. Feed rate.
4. Effective rake angle of tool.
5. Cutting speed.
6. type & quantity of cutting fluid.

* three different type of chips formed during metal cutting are as follow:-

1. Continuous chip:- A continuous chip is obtained when cutting ductile materials such as low carbon steel aluminium & copper. this chip is severely deformed & either comes off in the form of a long string, or curl into a tight roll as shown in fig.



⇒ Continuous chip.

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ASSIGNMENT SHEET

Branch: CIVIL ENGINEERING

Semester: B.Tech 7TH

Subject: FOUNDATION ENGINEERING

Code: D020713 (020)

- Q.1 a) Define Bearing capacity of a soil.
b) Give brief detail about Terzaghi theory.
- Q.2 a) What do you understand by U.B.C.?
b) Explain in detail about Assumption of Terzaghi.
- Q.3 a) What do you mean by S.B.C.?
b) Describe in brief about General shear failure.
- Q.4 a) What do you understand by Pile driver?
b) Explain the types of Pile foundation.
- Q.5 a) Define Negative skin Friction.
b) Explain Under reamed pile in a detail.
- Q.6 a) Define cast in situ concrete pile.
b) Explain in detail about pile driving Hammers.

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CHHATRAPATI SHIVAJI INSTITUTE OF TECHNOLOGY, DURG

DEPARTMENT OF MECHANICAL ENGINEERING

SESSION – APRIL-MAY - 2023

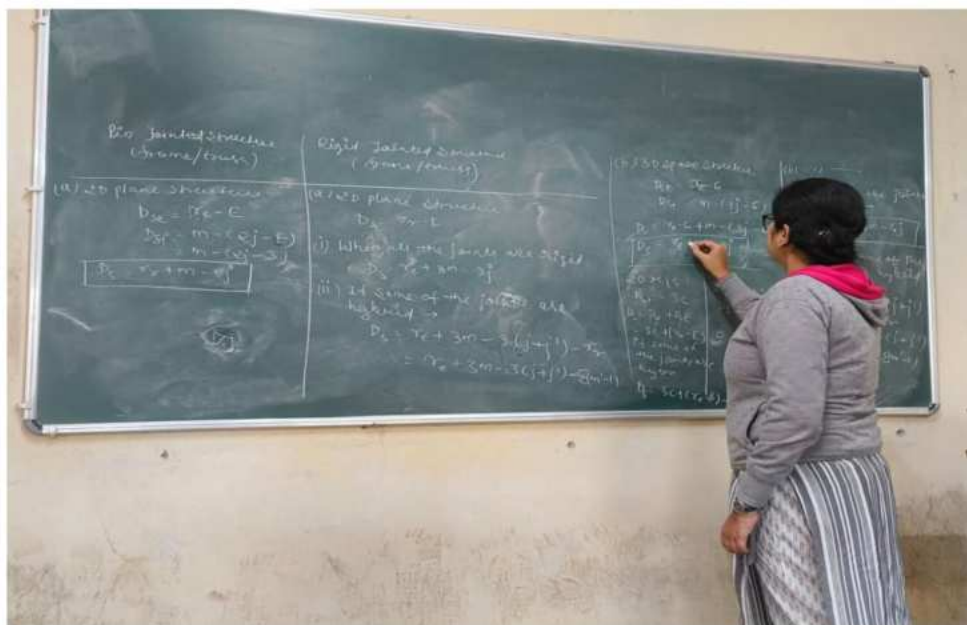
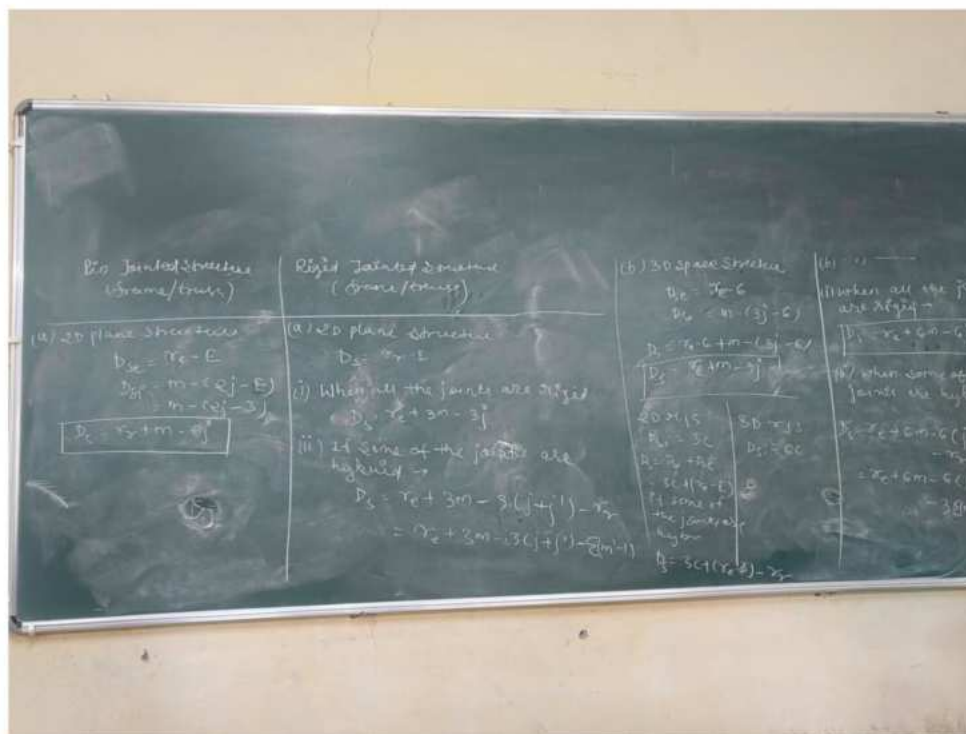
SUBJECT: - FLUID MECHANICS

SUBJECT CODE. : - B037412(037)

ASSIGNMENT UNIT 4

Unit-4

1. A pipe of diameter 20 cm and length 10^4 m is laid at a slope of 1 in 200. An oil of specific gravity 0.9 and viscosity 1.5 Poise is pumped up at the rate of 20 litres per second. Find the head lost due to friction. Also calculate the power required to pump the oil.
2. A pipeline of length 2000 m is used for power transmission. If 110.3625 kW power is to be transmitting through the pipe in which water having a pressure of 490.5 N/cm^2 at inlet is flowing, find the diameter of the pipe and efficiency of transmission if the pressure drop over the length of pipe is 98.1 N/cm^2 . Take $f = 0.0065$.
3. Prove that for a flow of viscous flow through a pipe: $\frac{U_{\max}}{\bar{u}} = 2.0$
where, $U_{\max}$ maximum velocity, and
 $\bar{u}$ average velocity
4. An oil of viscosity 0.1 Ns/m^2 and relative density 0.9 is flowing through a circular pipe of diameter of 50 mm and of length 300 m. The rate of flow of fluid through the pipe is 3.5 litres/s. Find the pressure drop in a length of 300 m.
5. Determine the wall shearing stress in a pipe of diameter 100 mm which carries water. The velocities at the pipe centre and 30 mm from the pipe centre 2 m/s and 1.5 m/s respectively. The flow in pipe is given as turbulent.



Picture of Chalk & Talk in Classroom



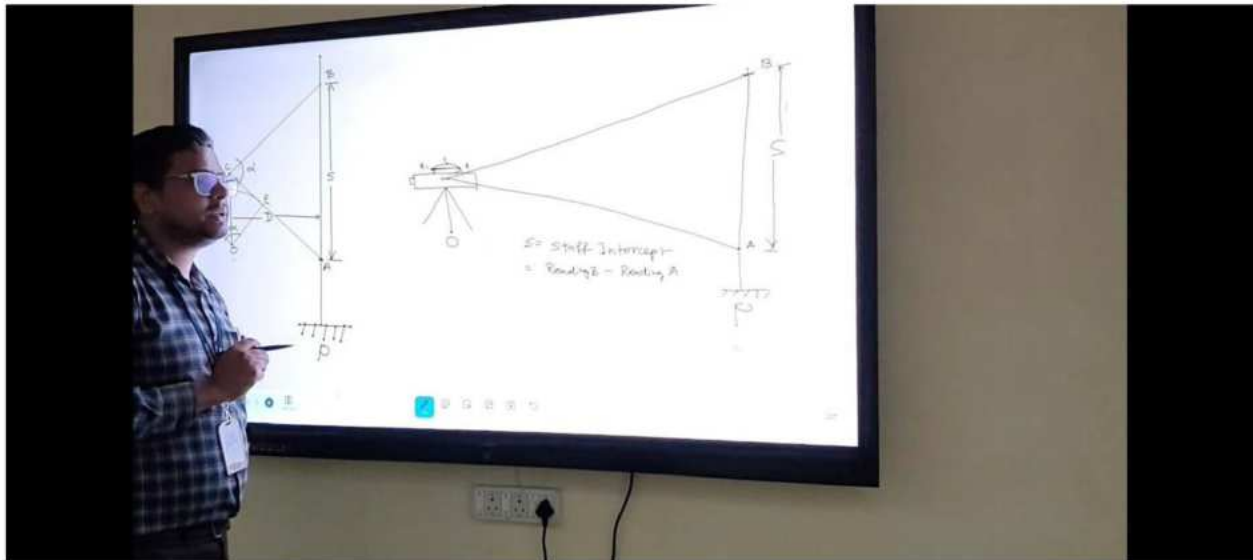
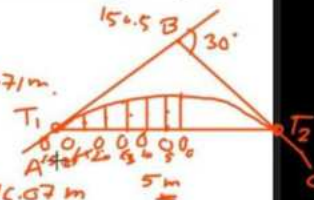
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Problem 1. Two tangents AB and BC intersect at a point B at 150.5 m chainage. Let us calculate all the necessary data for setting out a circular curve of 100 m radius and deflection angle 30° by the method of offsets from the long chord.

1. $TL = R \tan \frac{\phi}{2} = 26.79 \text{ m.}$
2. Chainage of $T_1 = 150.5 - 26.79 = 123.71 \text{ m.}$
3. Curve length $= \frac{\pi R \phi}{180} = 52.36 \text{ m.}$
4. Chainage of $T_2 = 123.71 + 52.36 = 176.07 \text{ m.}$
5. Length of long chord $C = 2R \sin \frac{\phi}{2} = 51.76 \text{ m.}$
6. Half Chord, $\frac{C}{2} = \frac{51.76}{2} = 25.88 \text{ m.}$
7. $O_0 = R - \sqrt{R^2 - \left(\frac{C}{2}\right)^2} = 9.41 \text{ m.}$

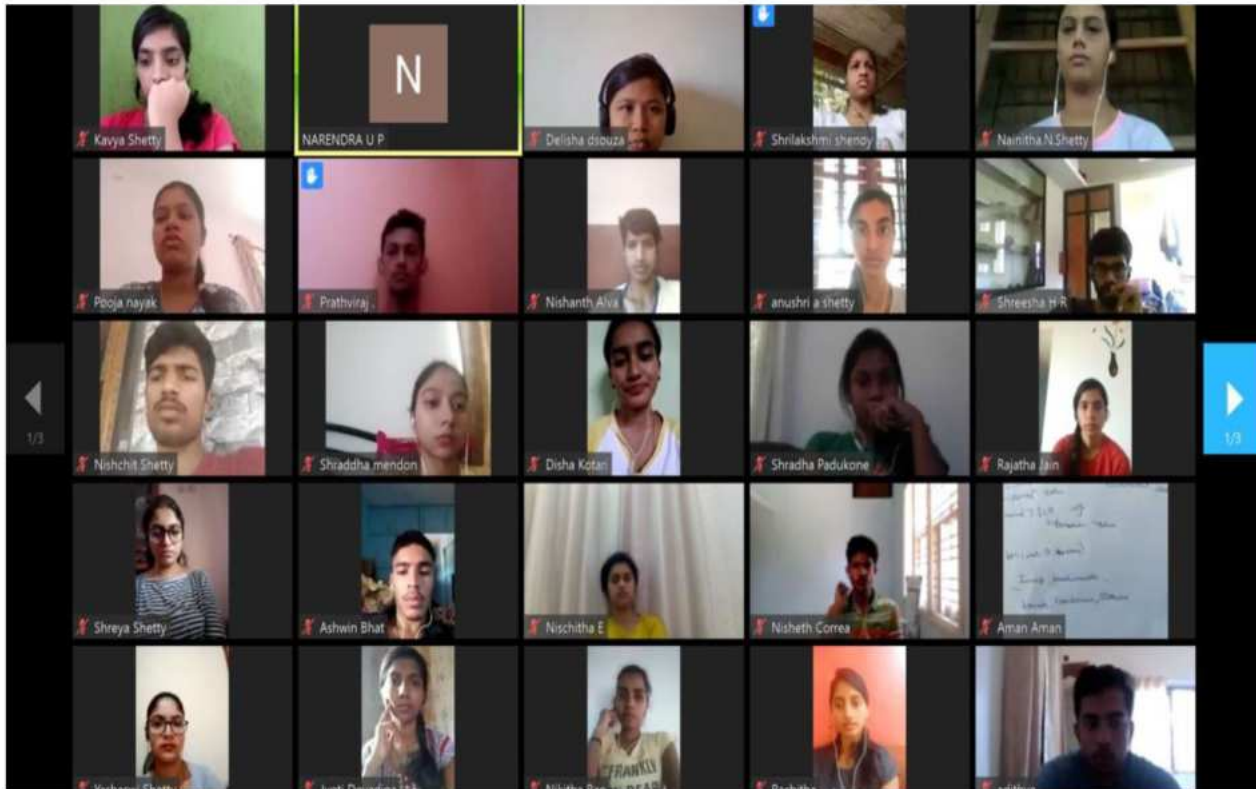


Picture of Online Teaching

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ONLINE TEACHING





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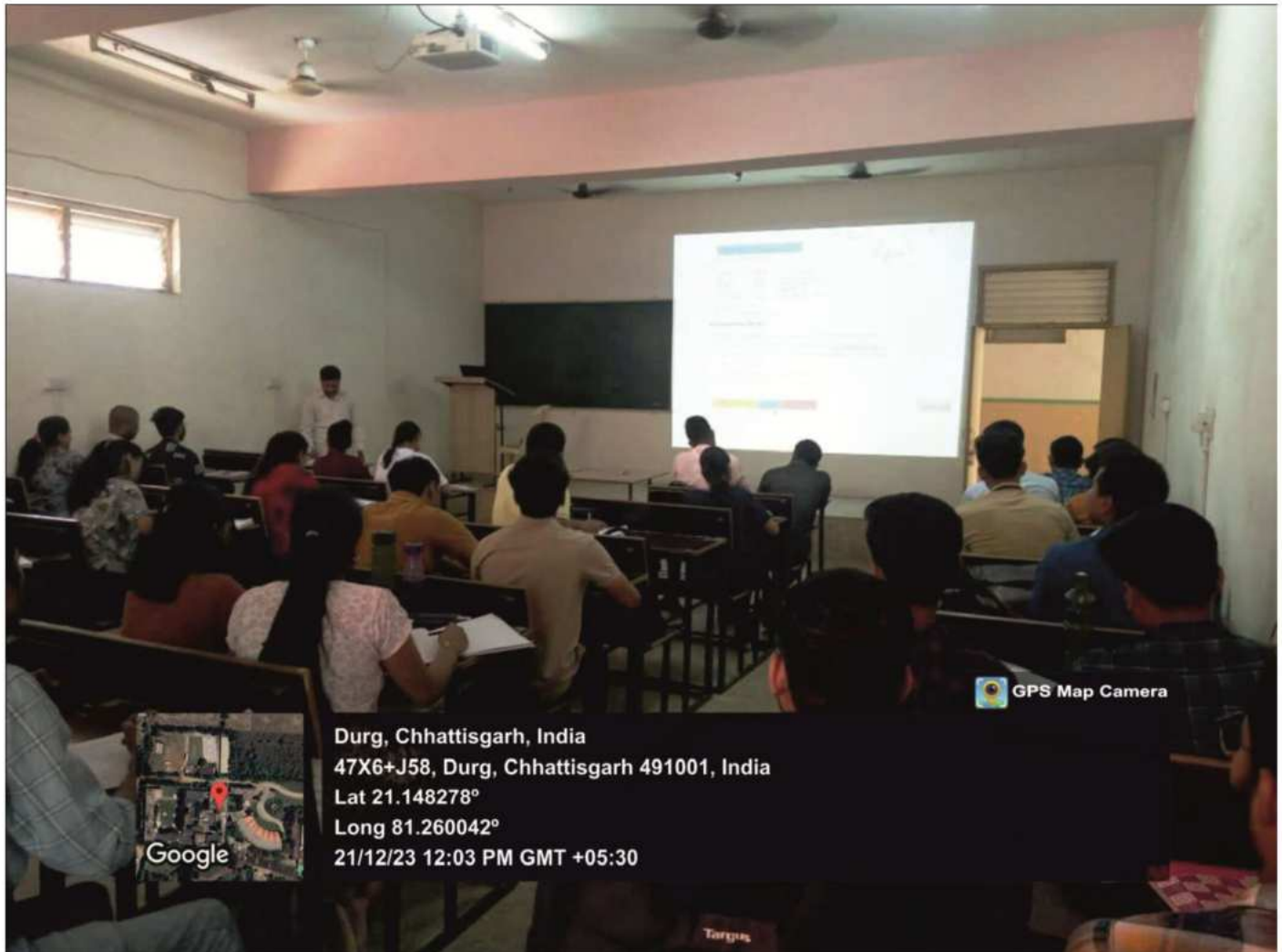


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E-CLASS ROOM WITH LCD PROJECTOR



 GPS Map Camera

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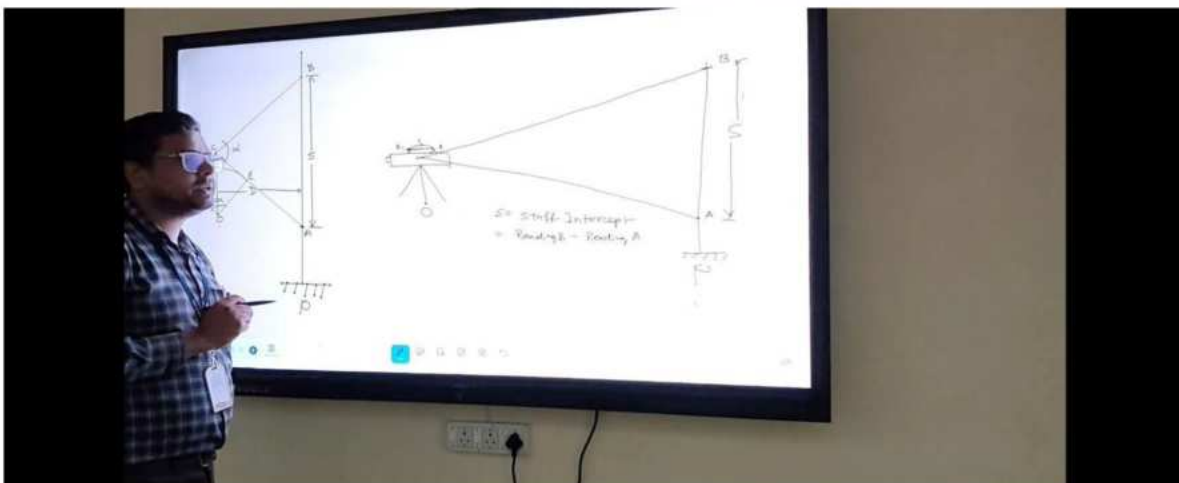
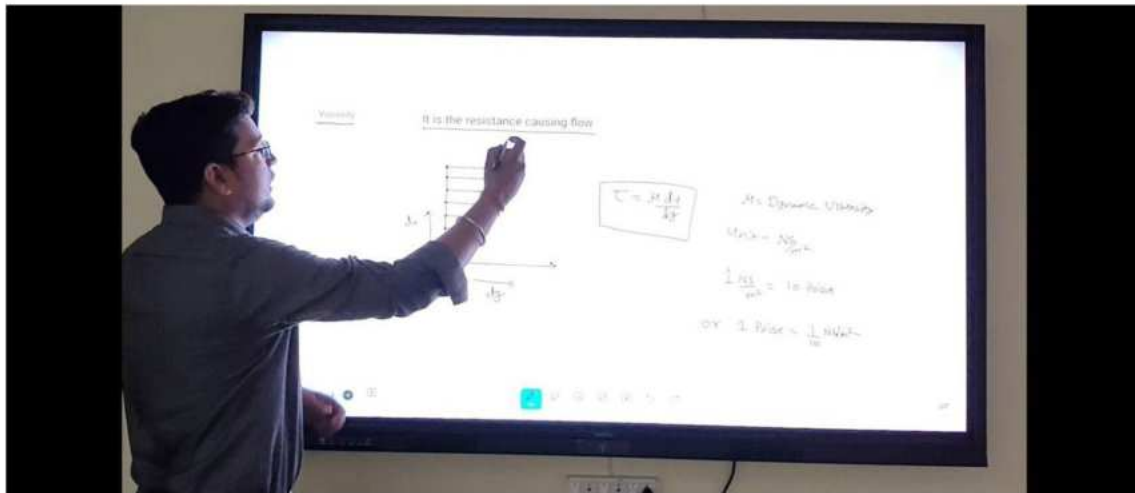
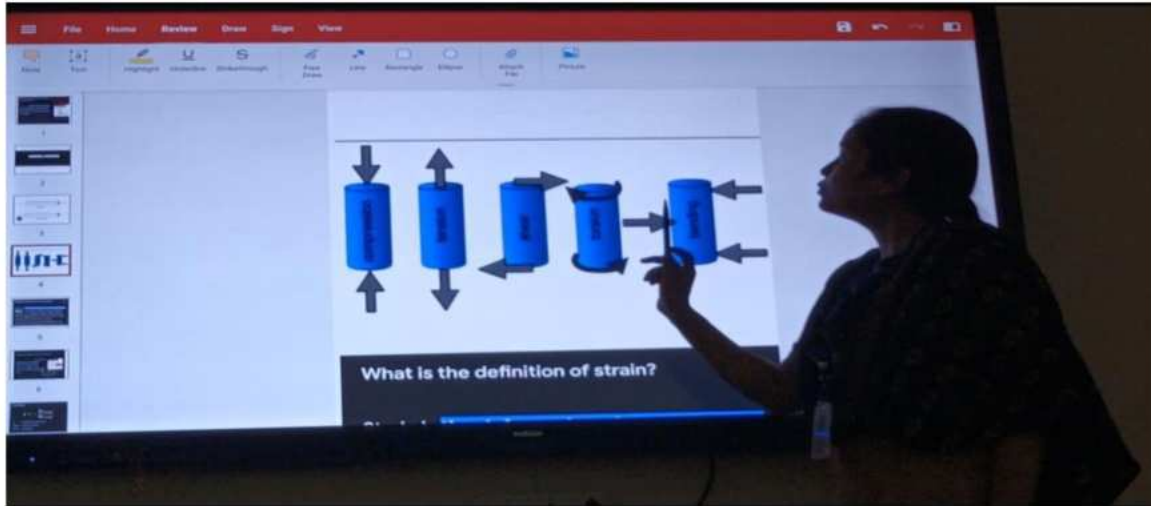
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Picture of E-Classroom with LCD Projector

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Department of Mechanical Engineering

Class Notes, Subject – Fluid Mechanics

Lecture-2

Unit-1

Syllabus-

Properties of fluid: Fluid, ideal and real fluid, properties of fluid: mass density, weight density, specific volume, specific gravity, viscosity, surface tension, capillarity, vapour pressure, compressibility and bulk modulus. Newtonian and non-Newtonian fluids
Fluid Statics: Pressure, Pascal's law, Hydrostatic law, Manometer, Hydrostatic force on submerged plane and curved surface, Buoyancy and Flotation.

Outcome/Importance/Implication-

1. List different types of fluid properties
2. Measurement of fluid pressure and its importance
3. Fluid pressure distribution in different types of submerged bodies like (Horizontal, Vertical, Curved)

Title -1

Properties of fluid: Fluid, ideal and real fluid,

Reference-

- *Fluid Mechanics & Hydraulics Machines-R.K.Bansal-Laxmi Publications.,Delhi (page no 1 to 6)*

Teaching Aids Used-

Chalk and Talk,

Video Link -1 –

<https://www.examfear.com/free-video-lesson/Class-11/Physics/Mechanical-Properties-of-Fluids/part-1.htm> ,

Prepared By: - Nishant Kumar Shukla
Assistant Professor, Department of Mechanical Engineering
Chhatrapati Shivaji Institute of Technology, Durg (Chhattisgarh)



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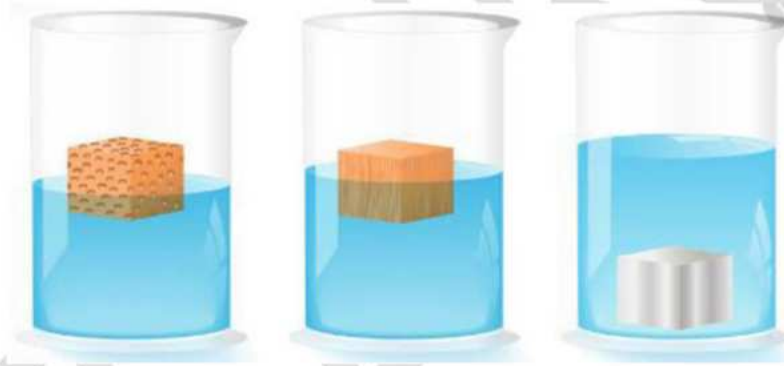
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where ρ is the density, m is the mass, and V is the volume. In some cases (for instance, in the United States oil and gas industry), density is loosely defined as its weight per unit volume, although this is scientifically inaccurate – this quantity is more specifically called specific weight.

⇒ *For a pure substance the density has the same numerical value as its mass concentration. Different materials usually have different densities, and density may be relevant to buoyancy, purity and packaging. Osmium and iridium are the densest known elements at standard conditions for temperature and pressure.*

Daily Life Example-

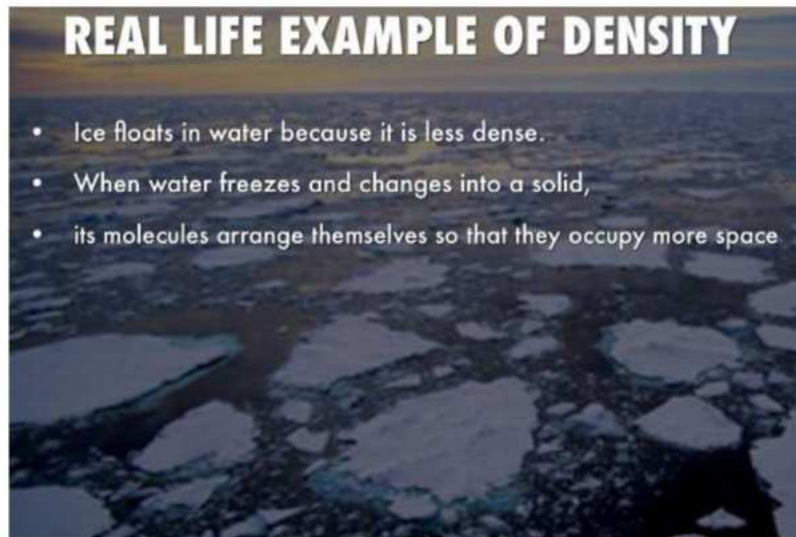
- Suppose a block of brass and a block of wood have exactly the same mass. If both blocks are dropped in a tank of water, why does the wood float and the brass sink? This occurs because the brass has a greater density than water, whereas the wood has a lower density than water.



- If you have ever tried to mix water and oil. Oil is less dense than water, so it floats on top. A boat also floats for the same reason, Its average density is less than that of water (because of the air in the hull) so it can also float.



Prepared By: - Nishant Kumar Shukla
Assistant Professor, Department of Mechanical Engineering
Chhatrapati Shivaji Institute of Technology, Durg (Chhattisgarh)



2. Specific weight or Weight Density :-

The specific weight, also known as the unit weight, is the weight per unit volume of a material.

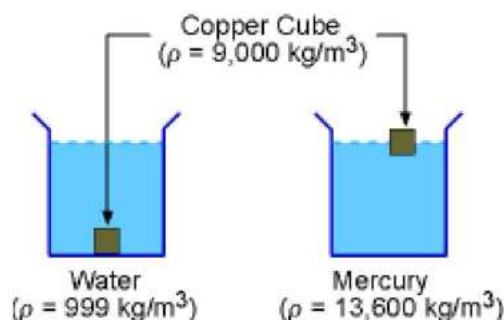
A commonly used value is the specific weight of water on Earth at 4°C, which is 9.807 kN/m³

The terms specific gravity, and less often specific weight, are also used for relative density. A common symbol for specific weight is γ , the Greek letter Gamma.

$$\gamma = \rho g$$

Unlike density, specific weight is not a fixed property of a material. It depends on the value of the gravitational acceleration, which varies with location. Pressure may also affect values, depending upon the bulk modulus of the material, but generally, at moderate pressures, has a less significant effect than the other factors.

Live Example-



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